12TH CHEMISTRY · CHAPTER 29 · CHAPTER TEST

Spectroscopy–1

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 29	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The IHD of benzene is:

(a) 1 (b) 2
(c) 3 (d) 4

2. UV-Vis spectroscopy uses the range:

(a) 1–10 nm (b) 200–800 nm
(c) 1–10 m (d) 10–50 nm

3. The transition needing least energy is:

(a) n→π* (b) σ→σ*
(c) π→π* (d) n→σ*

4. A C=O bond absorbs IR near:

(a) 3400 per cm (b) 2900 per cm
(c) 1700 per cm (d) 1100 per cm

5. One Cl atom gives M : M+2 peaks as:

(a) 1 : 1 (b) 3 : 1
(c) 9 : 6 (d) 1 : 2

6. The M+1 peak is due to:

(a) Bromine-81 (b) Chlorine-37
(c) Carbon-13 (d) Deuterium

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Define index of hydrogen deficiency.
ii. Differentiate emission and absorption spectra.
iii. Why are oxygen and hydrogen molecules IR inactive?
iv. How does conjugation affect the wavelength absorbed?
v. What is a molecular ion peak?
vi. Why does bromomethane give peaks at 94 and 96?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe the electronic transitions in UV-Vis spectroscopy.
(b) Calculate the atomic mass of boron (B-10 19.9%, B-11 80.1%).

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